Poornachandra G.

Bioinformatics & AI Engineering | Precision Medicine & Agentic AI Systems

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PROFESSIONAL SUMMARY

Bioinformatics Scientist and AI Systems Developer with 10+ years of experience in precision oncology, clinical genomics, and production-grade NGS and multi-omics pipelines. Combines deep expertise in biomarker discovery, variant interpretation, and translational research with hands-on development of AI-powered applications, RAG pipelines, and multi-agent systems. Experienced in reproducible workflow engineering, machine learning, and systematic AI evaluation, with a focus on biological relevance, scientific validity, and reliable, traceable computational workflows.

SELECTED HIGHLIGHTS & IMPACT

- Clinical NGS Pipelines: Built and productionized end-to-end NGS pipelines used in clinical and research workflows; significantly reduced sample turnaround time through automated QC and reporting.
- Variant Interpretation Workflows: Architected variant annotation and prioritization workflows (ACMG/AMP/ASCO/CAP) integrating multiple public and proprietary annotations to enable actionable clinical decision support.
- Biomarker Discovery & ML Initiatives: Led ML initiatives for biomarker discovery and patient stratification: feature engineering from multi-omics inputs, model validation, and deployment in production environments.
- Phenotype-Based Rare Disease Prioritization: Architected and built a tertiary analysis pipeline for phenotype-based gene prioritization, variant selection, and reporting in rare diseases. Presented this work and secured the top rank at the Bangalore Tech Summit (2025) as an independent researcher.
- Bioinformatics Team Mentorship & Hiring: Mentored and grew a bioinformatics team; led technical hiring, performance reviews, and cross-functional training.
- AI/LLM Engineering & Observability: Built multiple independent AI projects spanning LLM tooling, multi-agent frameworks, clinical AI apps, and developer infrastructure. Established AI evaluation and observability frameworks (Promptfoo, DeepEval, Ragas) for accuracy benchmarking, hallucination analysis, and retrieval quality validation.

SKILLS & TECHNOLOGIES

Pipeline Engineering	Nextflow • Benchmarking • Reproducibility • Seurat • GATK • Sentieon • DRAGEN • Exomiser • Franklin • NVIDIA Parabricks • FreeBayes • DeepVariant • DeepSomatic • PureCN • CNVKit • VEP • ANNOVAR • QC Automation • Kallisto • Salmon • DESeq2 • edgeR • Scanpy • GSEA
Programming & Engineering	Python • R • Git • CI/CD • Bash • REST APIs
Cloud, Platforms & Infrastructure	Docker • AWS (EC2, S3, Batch, HealthOmics) • Azure AI Foundry • DNAnexus • BaseSpace • Galaxy • GenePattern • Illumina Connected Analytics • Conda/Mamba • Jupyter • MLOps
Machine Learning & AI	Supervised/Unsupervised Learning • Feature Engineering • LightGBM • XGBoost • PyTorch • TensorFlow • Deep learning • Scikit-learn • PyCaret • AutoML • SHAP • Scipy • Pandas • NumPy • AlphaMissense • AlphaGenome • scGPT
LLMs & Agentic Systems	Fine-Tuning • LangChain • LangGraph • CrewAI • Multi-Agent Systems • RAG Pipelines • Prompt Engineering • Context Engineering • Quantization • Hugging Face • Transformers • Accelerate • Model Cards • PEFT (LoRA/QLoRA) • AI Evaluation & Observability (Promptfoo, DeepEval, Ragas, LangSmith, Langfuse) • AI Validation (LLM-as-a-Judge, Red Teaming, Hallucination Analysis) • LiteLLM • MCP (Model Context Protocol) • OpenAI / Anthropic / Gemini APIs • Ollama • W&B • MLflow
Applied Bioinformatics & Clinical-omics	WGS • WES • TES • RNA-Seq • scRNA-Seq • Spatial Omics • Liquid Biopsy • Pharmacogenomics • Companion Diagnostics • Variant Interpretation (ACMG/AMP/ASCO/CAP) • MSI • HRD • TMB • CNV • SV • Rare Disease Genomics • Phenotype-based Gene Prioritization
Databases & Knowledgebases	TCGA • GDC • GENIE • cBioPortal • COSMIC • CMP • OncoKB • CIViC • DepMap/CCLE • GDSC • CGI • Open Targets • ClinVar • ClinGen • OMIM • HPO • Orphanet • DECIPHER • GenCC • HGMD • gnomAD • GTEx • CellxGene • HTAN • HuBMAP • UniProt • dbGaP • dbNSFP
Visualization & UI	Streamlit • Plotly • Matplotlib • Seaborn
Genai Developer Tools	Claude Code • Codex • Antigravity • GitHub Copilot • Cursor • Windsurf • Continue • Hermes • PI • Openclaw • Opencode • VS Code • paperclip

Leadership & Delivery	Team Building • Technical Hiring • Mentoring • Technical Roadmapping • Stakeholder Communication • HIPAA Compliance • Agile / Scrum • Risk Management • JIRA • Cross-Functional Training

PROFESSIONAL EXPERIENCE

Cellworks - Lead Scientist Bioinformatics
2024 - Present

- Led the Bioinformatics R&D team: defined technical roadmap, recruited and mentored scientists, drove project prioritization, and aligned strategy with executive leadership.
- Partnered with executive leadership to align strategy and communicate progress, while providing team members with the broader context needed to execute effectively.
- Built multi-stage annotation and prioritization pipelines applying AMP/ASCO/CAP guidelines across public and proprietary annotation sources.
- Used AlphaGenome to identify candidate TERT promoter variants with potentially significant effects on gene expression and functional relevance in cancer.
- Designed and implemented input ingestion, monitoring, and reporting frameworks that reduced manual review time by 20%.
- Established risk management practices for bioinformatics pipelines - identified potential failure points, data-quality risks, and downstream repercussions, and implemented mitigation strategies.
- Designed structured upskilling sessions leveraging supplemental projects beyond core deliverables, improving engagement, critical thinking, and problem-solving across team members.
- Leveraged technical competitive intelligence and published methodologies to develop and benchmark an in-house bioinformatics pipeline inspired by Foundation Medicine (FM).

Cellworks - Senior Scientist, Reporting Manager
2021 - 2024

- Developed comprehensive workflows for microsatellite instability (MSI), homologous recombination deficiency (HRD), mutational signatures, tumor mutational burden (TMB), and indel burden.
- Built high-throughput variant calling stacks on AWS following GATK best practices, including mutation and CNV calling, and biomarker extraction.
- Built and led the bioinformatics team: recruited, trained, mentored, and reviewed the work of team members; created custom training protocols that accelerated onboarding and competency development.
- Designed input-agnostic workflows supporting multiple formats (FASTQ, BAM, VCF, HDF5, structured reports) across diverse commercial vendors (4BC, Tempus, Caris, Foundation Medicine, Alacris) and research workflows.
- Built cancer driver and copy number variation (CNV) vs. mRNA expression correlation databases for 30+ cancer types using public datasets and computational tools.
- Developed automated multi-omics interpretation pipelines for high-impact clinical trial datasets from biopharma partners including Takeda and Genentech.
- Created an in-house Variant of Uncertain Significance (VUS) reclassification pipeline using an Ensembl-based approach, improving classification accuracy by >40%.
- Analyzed single-cell RNA-seq data from Human Protein Atlas normal-tissue samples with Seurat, Scanpy, and limma to establish in-silico control baselines for distinguishing cancer-specific expression signatures.

Cellworks - Senior Scientist, Bioinformatics
2018 - 2021

- Developed and benchmarked secondary analysis pipelines (Illumina DRAGEN, NVIDIA Parabricks, Sentieon), optimized for turnaround time, resource cost, and variant calling accuracy.
- Led custom panel design for liquid-biopsy samples; collaborated with Cleveland Clinic Florida to build a machine learning pipeline using longitudinal genomic data to predict early relapse in Melanoma.
- Built an analytical framework to process pediatric cancer cases; served as primary domain owner for pediatric hematological malignancies.
- Designed and implemented a Panel of Normals (PoN) workflow for CNV and mutation calling, improving support for tumor-only clinical analyses.
- Created multiple tertiary analysis workflows (GSEA, pathway enrichment, PCA, statistical models, ML pipelines) for biomarker discovery and treatment-response characterization.
- Developed multi-omics interpretation pipelines for the Foundation Medicine clinical genomic dataset (18,000+ patient cases).
- Compiled a comprehensive diagnostic assay and genomic test inventory across 88 cancer types based on NCCN, FDA, and international clinical guidelines.

Cellworks - Scientist, Clinical Translation
2017 - 2018

- Curated therapeutic rationale libraries to support computational biology predictions, reducing turnaround time for live clinical case analyses by >40%.
- Analyzed 200+ complex cancer cases, gaining deep expertise in target selection using biological network models and drug interaction mechanisms in precision oncology.

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- Conducted rigorous benchmarking of external sequencing pipelines (DNAnexus, BeOne, MedGenome) to establish analytical validity, confidence, and cross-platform reproducibility.
- Developed an in-house somatic variant annotation and filtering pipeline with custom tier classification, eliminating recurring external software fees and improving clinical interpretability.

Cellworks - Associate Scientist 2015 - 2017

- Processed, interpreted, and harmonized multi-omics outputs (karyotyping, aCGH, IHC, somatic variants) for >500 cases, translating insights into automated data-harmonization ETL pipelines.
- Curated and automated retrospective in-house dataset creation by integrating patient and cell-line data for >10,000 cases from various sources like COSMIC, TCGA, cBioportal, GEO, SRA and published clinical studies.
- Standardized genomic data interpretation from external medical partners including Columbia University, University of Florida, Cedars-Sinai, Mayo Clinic, and Washington University.
- Automated report-generation workflows using reproducible pipelines and templates, reducing preparation time from ~15 hours to just minutes per week.

KEY PROJECTS & INNOVATIONS

Portal Intelligence LangGraph • LangSmith • Multi-agents • Embeddings • RBAC • DeepEval

- Designed multi-agent orchestration workflows for vendor order management, coordinating specialized agents, external tools, and contextual memory with human-in-the-loop validation at critical decision points. Designed and maintained golden datasets, benchmark queries, and expected-answer sets that powered repeatable evaluation of retrieval quality, response accuracy, hallucination reduction, and regression testing.

JobAIder

- Built an AI-driven career platform integrating resume and Obsidian vault parsing, job description analysis, profile-to-role gap assessment, personalized upskilling plans, ATS-optimized resume generation, and role-specific mock interviews.

HealthForge LLM API • context management • full-stack • workflow automation

- Designed and built an AI-powered health planning platform integrating personalized health profiles, context-aware LLM interactions, wellness plan generation, and WhatsApp reminders into an end-to-end application.

EDUCATION

B.Tech in Biotechnology — JNTUA College of Engineering Pulivendula (JNTUA CEP) 2015

HONORS & AWARDS

- **Bangalore Tech Summit 2025 Top Rank (Independent Research) (2025):** Awarded Top Rank at Bangalore Tech Summit 2025 as an independent researcher for architecting a phenotype-based gene prioritization and tertiary analysis pipeline for rare disease diagnostics.

SELECTED PUBLICATIONS

- Co-authored 9 peer-reviewed oncology publications and conference abstracts in J Neurooncol, J Clin Oncol, and Blood/ASCO, covering computational biomarker modeling, AML/MDS biosimulation, and therapy-response modeling. Available on ORCID: 0009-0009-1610-1205

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